Human Aspects of Machine Learning Research Group

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A Unifying Post-Processing Framework for Multi-Objective Learn-to-Defer Problem

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Abstract

Learn-to-Defer is a paradigm that enables learning algorithms to work not in isolation, but as a team with human experts. In this paradigm, we permit the system to defer a subset of its tasks to the expert. Although there are currently systems that follow this paradigm and are designed to optimize the accuracy of the final human-AI team, the general methodology for designing such systems under a set of constraints (e.g., algorithmic fairness, expert intervention budget, defer of anomaly, etc.) remains largely unexplored. In this paper, using a d-dimensional generalization to the fundamental lemma of Neyman and Pearson (d-GNP), we obtain the Bayes optimal solution for learn-to-defer systems under a variety of constraints. Furthermore, we design a generalizable algorithm to estimate that solution, and apply this algorithm to COMPAS and ACSIncome dataset. Our algorithm shows improvements in terms constraint violation over a set of baselines.

Deferral Learning with Limited Budget

- **Setting:**
  - Human and machine both make mistakes for classifying a datapoint
  - Minimize mistakes by learning when to defer
  - We have secondary objectives to keep bounded
- **Formal Setting:**

$$h^*, r^* \in \underset{(h,r)\in\mathcal{H}\times\mathcal{R}}{\operatorname{argmin}} \; \mathbb{E}[\ell_{AI}(h(X),Y)\mathbb{1}_{r(X)=0} + \ell_H(M,Y)\mathbb{1}_{r(X)=1}],$$
$$s.t. \quad \mathbb{E}[\ell_c^i(X,h(X),Y)] \le \delta_i \text{ for } i \in \{1,\dots,m\}$$

  - h and r are classifier and deferral, M is the human decision for a feature X and the true label is Y
- Joint Learning of h and r: e.g., fairness criteria aren't compositional (Yule's effect)
- Regularization: computational inefficiency, and discontinuous loss

Hardness and Optimality

- **Complexity of Optimal Deterministic Rules:** Let the losses be $0-1$ loss for human and the classifier and assume $\mathcal{X}$ to be finite. Finding an optimal deterministic deferral rule for a bounded human intervention budget is equivalent to 0-1 Knapsack problem and therefore an NP-Hard problem.
- **Random Relaxation:** Equivalence to a a linear functional programming

$$f^* = [f_1^*, \dots, f_{K+1}^*] \in \operatorname{argmax}_{f\in\Delta_{K+1}^{\mathcal{X}}} \mathbb{E}_X\big[\langle f(X), \psi_{m+1}(X)\rangle\big]$$
$$\text{s.t.} \quad \mathbb{E}_X\big[\langle f(x), \psi_i(x)\rangle\big] \le \delta_i \text{ for } i \in \{1,\dots,m\}$$

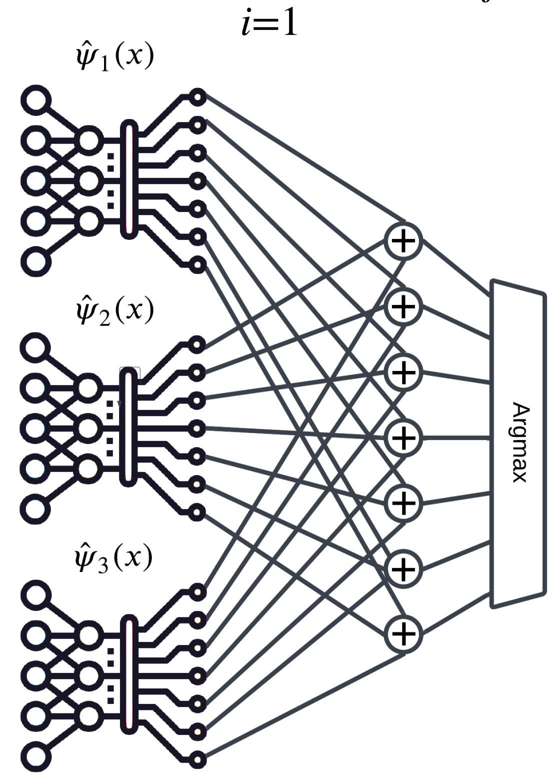
A Generalization to Neyman-Pearson Lemma

- **Neyman-Pearson Lemma:** Most-powerful test for a size at most δ is a likelihood-ratio test, i.e.,

$$T^* = \operatorname{argmax}_T \mathbb{E}_Q[T(X)] \text{ s.t. } \mathbb{E}_P[T(X)] \le \delta$$

where $T^*(x) = 1$ where $Q(x) > kP(x)$ and $T^*(x) = 0$ where $Q(x) < kP(x)$

- **Theorem (informal):** If bounds of constraints are interior points of all possible pairs of constraints, when there is a *single maximizer*, then $f^*(x) = \operatorname{argmax}_j \big[\psi_{m+1}(x) - \sum_{i=1}^m k_i\psi_i(x)\big]_j$ and if not, then there is a probability mass over the maximizers, and if we know that constraints are achieved tightly. All optimal solutions to the linear functional programming is of form above.
- **Theorem (informal):** In case of a single constraint, k_1 is the root of a monotone function with known closed-form, and a random predictor is drawn for the cases that we don't have a *single maximizer*

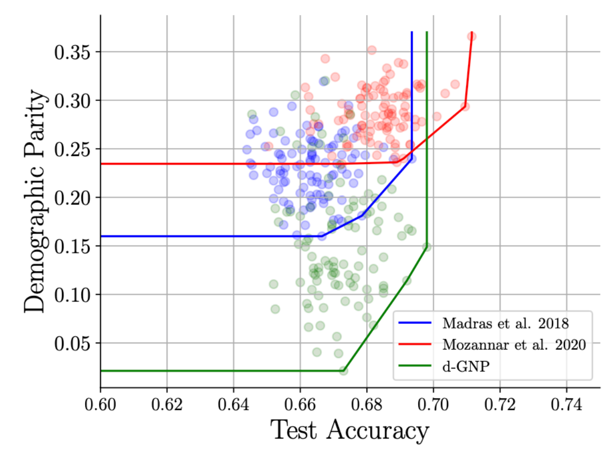


- **Constraint and Objective Generalization**: If γ measures the sensitivity of the constraint to the change of prediction then it generalizes with $O(\sqrt{\log n/n}, \sqrt{\log(1/\epsilon)/n}, \epsilon')$ and $O((\log n/n)^{1/2\gamma}, (\log(1/\epsilon)/n)^{1/2\gamma}, \epsilon')$, respectively

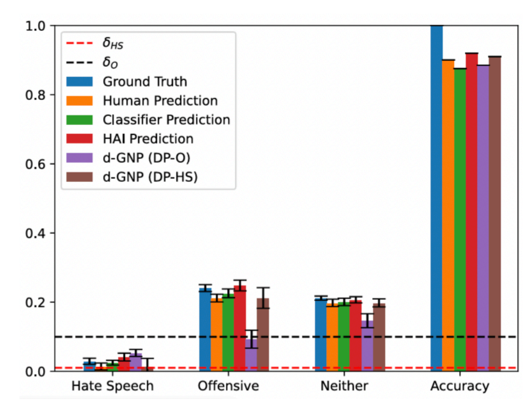
Type of Constraint	Embedding Function $\psi(x)$
Accuracy	$[P(Y=1\mid X=x), \dots, P(Y=K\mid X=x), P(Y=M\mid X=x)]$
Expert Intervention Budget	$[0,\dots,0,1]$
OOD Detection	$[0,\dots,0,\frac{f_X^{\text{out}}(x)}{f_X^{\text{in}}(x)}]$
Demographic Parity	$(\frac{\mathbb{1}_{A=1}}{Pr(A=1)} - \frac{\mathbb{1}_{A=0}}{Pr(A=0)})[0,1,\Pr(M=1\mid x)]$
Equality of Opportunity	$(\frac{\mathbb{1}_{A=1}}{Pr(Y=1,A=1)} - \frac{\mathbb{1}_{A=0}}{Pr(Y=1,A=0)})[0,\Pr(Y=1\mid x),\Pr(M=1,Y=1\mid x)]$

Experiments

- **COMPAS Dataset**



- **Hatespeech Dataset**



- **ACSIncome**